Proof Tree Nesting Depth Test

Test 1 : Simple (depth 1)

$$\dfrac{\ulcorner p \urcorner^{[1]} \quad \dfrac{}{q}\,[\text{axiom}]}{p \Rightarrow q}\,[\Rightarrow\text{-intro}^{[1]}]$$

Test 2 : Moderate (depth 2)

$$\dfrac{\ulcorner p \urcorner^{[1]} \quad \dfrac{\ulcorner q \urcorner^{[2]} \quad \dfrac{}{r}\,[\text{axiom}]}{q \Rightarrow r}\,[\Rightarrow\text{-intro}^{[2]}]}{p \Rightarrow (q \Rightarrow r)}\,[\Rightarrow\text{-intro}^{[1]}]$$

Test 3 : Deep (depth 3)

$$\dfrac{\ulcorner p \urcorner^{[1]} \quad \dfrac{\ulcorner q \urcorner^{[2]} \quad \dfrac{\ulcorner r \urcorner^{[3]} \quad \dfrac{}{s}\,[\text{axiom}]}{r \Rightarrow s}\,[\Rightarrow\text{-intro}^{[3]}]}{q \Rightarrow (r \Rightarrow s)}\,[\Rightarrow\text{-intro}^{[2]}]}{p \Rightarrow (q \Rightarrow (r \Rightarrow s))}\,[\Rightarrow\text{-intro}^{[1]}]$$

Test 4 : Very Deep (depth 4)

$$\dfrac{\ulcorner p \urcorner^{[1]} \quad \dfrac{\ulcorner q \urcorner^{[2]} \quad \dfrac{\ulcorner r \urcorner^{[3]} \quad \dfrac{\ulcorner s \urcorner^{[4]} \quad \dfrac{}{t}\,[\text{axiom}]}{s \Rightarrow t}\,[\Rightarrow\text{-intro}^{[4]}]}{r \Rightarrow (s \Rightarrow t)}\,[\Rightarrow\text{-intro}^{[3]}]}{q \Rightarrow (r \Rightarrow (s \Rightarrow t))}\,[\Rightarrow\text{-intro}^{[2]}]}{p \Rightarrow (q \Rightarrow (r \Rightarrow (s \Rightarrow t)))}\,[\Rightarrow\text{-intro}^{[1]}]$$

Test 5 : Horizontal siblings (depth 2)

$$\dfrac{\ulcorner p \wedge q \urcorner^{[1]} \quad \dfrac{\ulcorner p \urcorner^{[1]} \quad \ulcorner q \urcorner^{[1]}}{r}\,[\wedge\text{-elim}]}{(p \wedge q) \Rightarrow r}\,[\Rightarrow\text{-intro}^{[1]}]$$